

- Q.27 What do you mean by wireless propagation link budget.
- Q.28 Draw the schematic or cordless system, explain in brief its working.
- Q.29 What is sectoring? Explain its need.
- Q.30 Explain concept of Frequency Reuse.
- Q.31 Explain the concept of showing margin.
- Q.32 Define Adjacent channel interference; show how it is minimized.
- Q.33 Differentiate between TDMA and CDMA in mobile communication
- Q.34 Explain in brief LTE
- Q.35 Explain the concept of OFDM.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Draw and explain the architecture of GSM system.
- Q.37 Compare 2G/3G/4G/5G systems of wireless communication.
- Q.38 Explain the concept of Antenna diversity and wireless channel capacity.

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3rd Sem / Mechatronics Subject:- Mobile & Wireless Communication

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Expand CDMA
- carrier division multiple access
 - code division multiple access
 - Curve division multiple access
 - None of these
- Q.2 MSC stands for
- mobile space center
 - mobile switching center
 - mobile switched code
 - none of these
- Q.3 Expand DTE
- data tone equipment
 - data terminal equipment
 - data time equipment
 - none of these
- Q.4 SMS stands for
- start messaging service
 - short messaging
 - subtle message service
 - none of these

- Q.5 MMS stands for
- a) mobile messaging service
 - b) multimedia messaging service
 - c) medium messaging service
 - d) none of these
- Q.6 UHF frequency range:
- a) user high frequency
 - b) ultimate high frequency
 - c) ultra-high frequency
 - d) none of these
- Q.7 Expand IMEI
- a) international mobile equipment identity
 - b) internal mobile equipment identity
 - c) internet mobile equipment identity
 - d) none of these
- Q.8 Expand GSM
- a) global system for mobile
 - b) general system for mobile
 - c) gross system for mobile
 - d) none of these
- Q.9 What is the shape of the cell in mobile communication
- a) circular
 - b) hexagon
 - c) pentagon
 - d) any shape
- Q.10 Expand GPRS
- a) global packet radio service
 - b) general packet radio service
 - c) geometric packet radio service
 - d) none of these

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Explain Multipath Fading.
- Q.12 Explain MIMO.
- Q.13 What is the shape of cell in the cellular system?
- Q.14 Explain adjacent channel interference.
- Q.15 GPS_____
- Q.16 What is Hand Off? Explain.
- Q.17 What is Roaming? Explain.
- Q.18 Elaborate Wi-Fi.
- Q.19 Who invented wireless communication.
- Q.20 What is the relation between frequency and wavelength?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain Shadowing in wireless communication.
- Q.22 Explain the concept of fading margin.
- Q.23 What is spread spectrum multiple access (SSMA) technique?
- Q.24 What is Time Division Multiple Access? Explain with the help of suitable diagram.
- Q.25 What is bluetooth? Write its applications
- Q.26 What are the limitations of First generations (1G) cellular systems?